Class05

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#Lab 4 #For the installation of GGplot we need to do: install.packages("ggplot2")

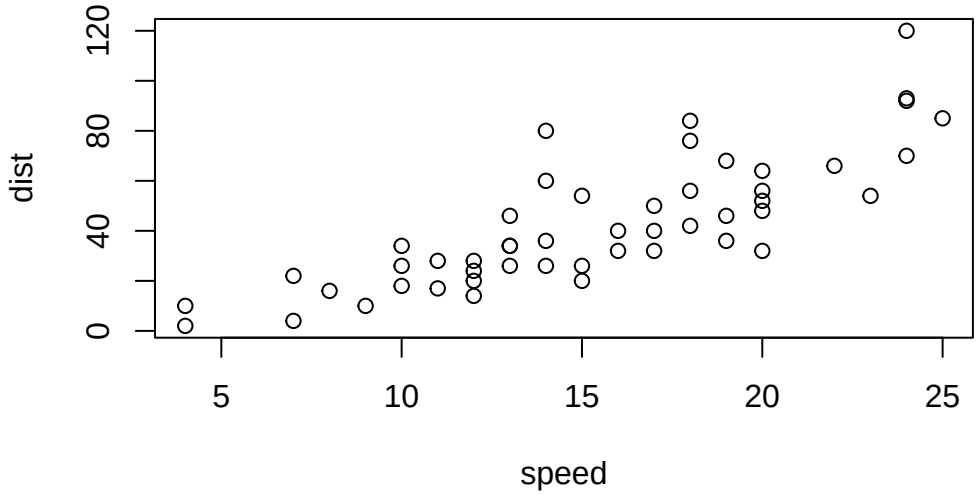
Anytime I need to use this package I need

```
library(ggplot2)
```

View(cars)

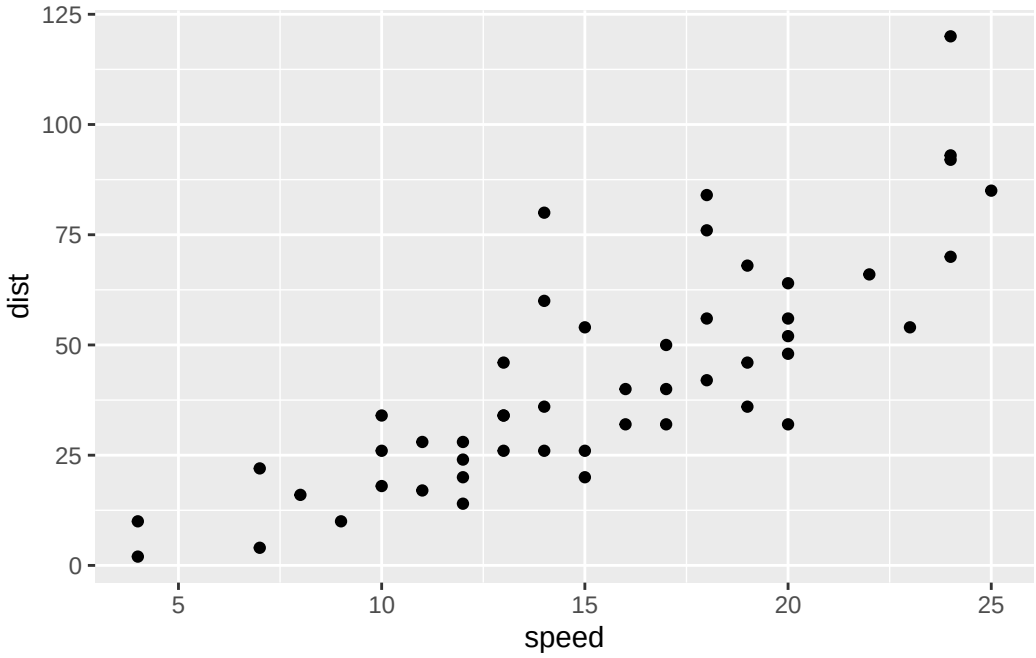
A quick base R plot- this is not ggplot.

```
plot(cars)
```

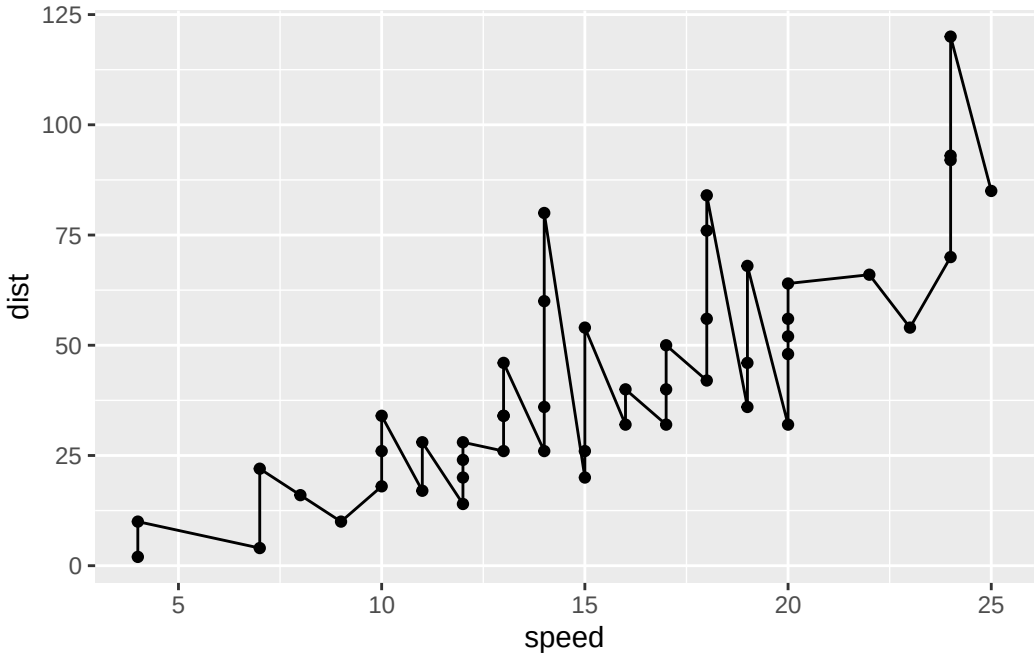


#First ggplot (data + aesthetic + geom)

```r
ggplot(data = cars) + aes(x = speed ,y = dist) + geom_point()
```

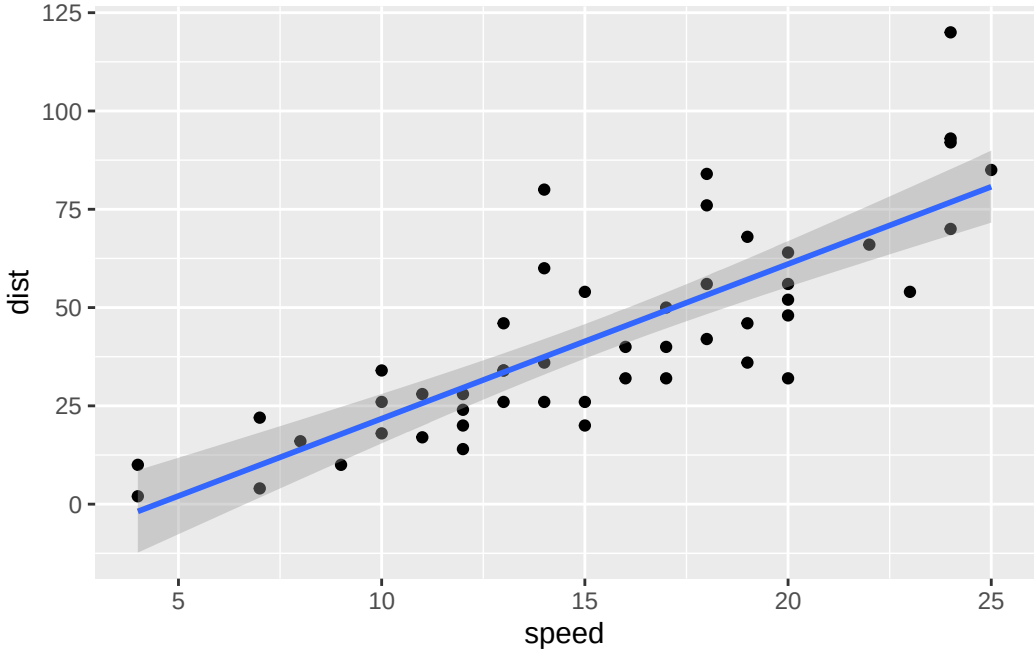



```r
p <- ggplot(data = cars) + aes(x = speed ,y = dist) + geom_point()

#Add a line geom with geom_line()
p + geom_line()
```




```r
p + geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



```r
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

Analysis of the dataset:

```r
g <- ggplot(data = genes) + aes(x= Condition1, y = Condition2, col = State) + geom_point()

#Number of genes
nrow(genes)
```

```
[1] 5196
```

```r
#Names of columns
colnames(genes)
```

```
[1] "Gene"       "Condition1" "Condition2" "State"
```

```r
#Number of columans
ncol(genes)
```

```
[1] 4
```

```r
#Summary of the State column in genes:
table(genes$State)
```

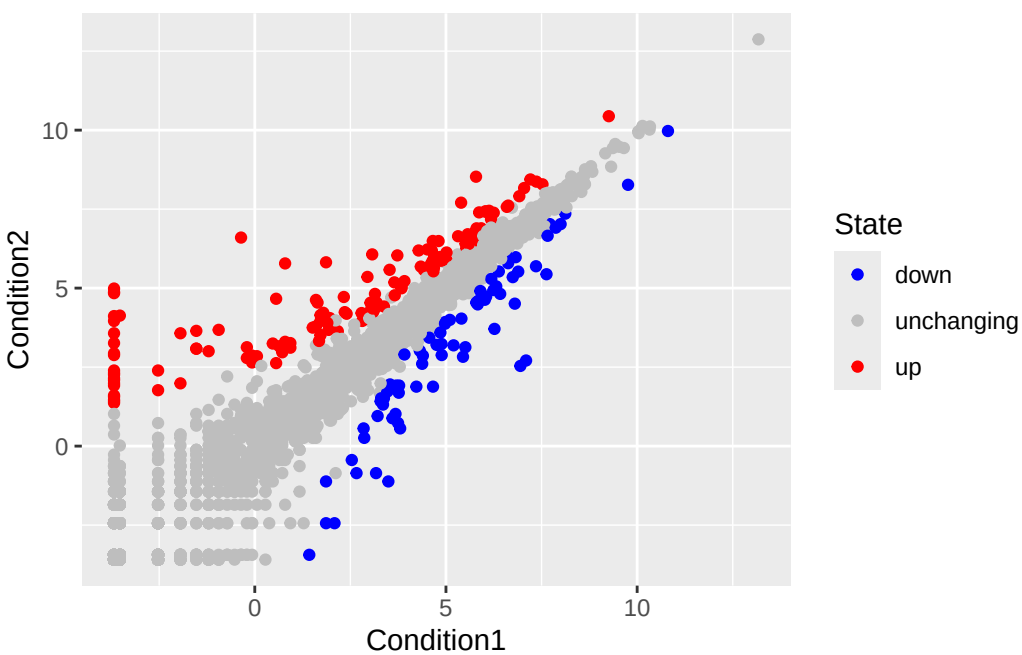
```
      down unchanging         up
        72       4997        127
```

```
#upregulated genes are 127

#We look at the up section for the value for it.

#Fraction of up genes:
round((table(genes$State) / nrow(genes)) * 100, 2)
```

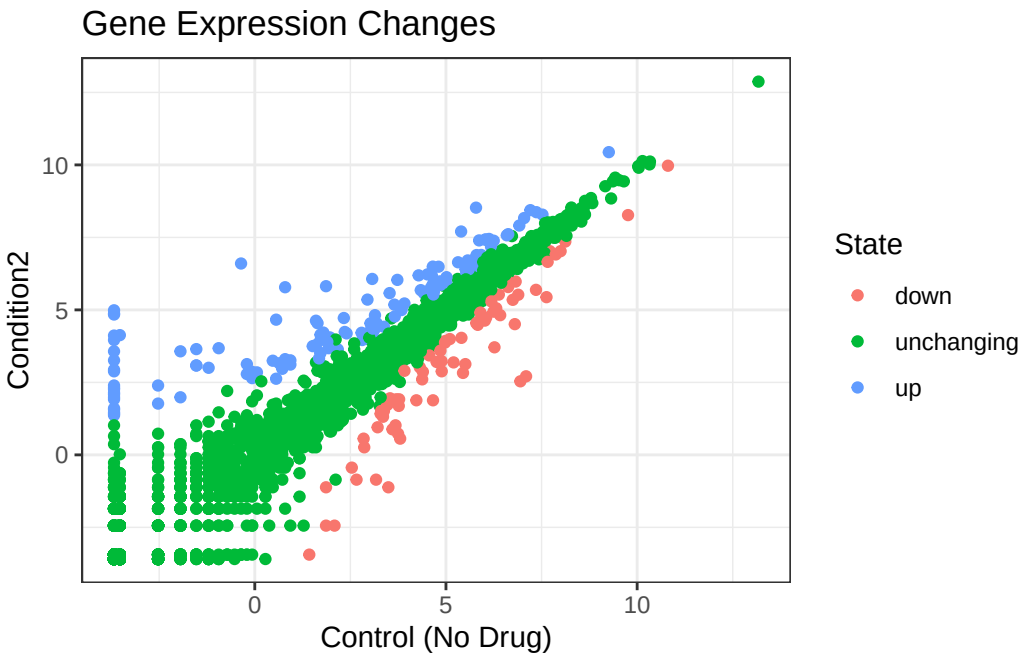
```
      down unchanging         up
      1.39      96.17       2.44
```

Color to ggplot

```
g + scale_color_manual(values = c("blue", "grey", "red"))
```




```
g+ labs(title = "Gene Expression Changes", x = "Control (No Drug)") + theme_bw()
```


Gene Expression Changes
Condition2
10
5
0
0
5
10
Control (No Drug)
State
down
unchanging
up